

Summary

AI Robotics Engineer with an M.S. in Electrical Engineering, focused on ROS 2, Isaac Sim / Isaac Lab, MoveIt 2, cuMotion, visual servoing, and physical-robot validation. Experienced in connecting perception, force-guided contact search, robot learning, policy inference, and Dockerized robotics software into practical manipulation workflows.

Key Results

3-4x imitation-learning / GROOT-VLA training-data throughput	70% → 90% cable-insertion success in 10 trials
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Experience

AMTRAN TECHNOLOGY CO., LTD. | AI Robotics Engineer May 2025 - Present | New Taipei, Taiwan

- Built ROS 2 control and simulation workflows for industrial and collaborative robot platforms with Isaac Sim / Isaac Lab, MoveIt 2, and cuMotion, covering motion planning, collision handling, pick-and-place, and cable tasks.
- Delivered Isaac GROOT N1.7 data conversion, model training, and policy inference, separately validating virtual-data training followed by real-world inference and real-data training/inference workflows.
- Built a two-part ROS 2 / WebSocket training-data collection flow that automated robot-arm tasks and recorded actions, images, robot state/actions, and timestamps, reaching 3-4x manual throughput and reducing manual staffing needs by two operators.
- Developed YOLO OBB / OpenCV visual servoing for HDMI insertion and AprilTag / OpenCV connector localization for cable unplugging; integrated wrist-mounted six-axis force/torque sensing and force-guided contact search into final insertion recovery with bounded spiral micro-search. Validated on the physical robot; incremental success-rate impact is not yet quantified.

ISCOM | Software Engineer Intern Feb 2022 - Jun 2022 | Kaohsiung, Taiwan

- Participated in backend API development, frontend integration, and web feature implementation using C#, ASP.NET MVC, JavaScript, jQuery, MS SQL, and Git.

Skills

Robot Control & Validation	ROS 2, Isaac Sim / Isaac Lab, MoveIt 2, cuMotion, Visual Servoing, physical-robot validation
Physical AI & Robot Learning	Isaac GROOT N1.7, VLA, Policy Inference, Imitation Learning, Robot State / Action
Perception & Force-Guided Insertion	YOLO OBB + OpenCV, AprilTag, RealSense, Hand-Eye Calibration, six-axis force/torque sensing, force-guided contact search
Robotics Software & Integration	Python, C++, FastAPI, React / TypeScript, WebSocket, Docker, Linux

Education & Honors

- M.S. in Electrical Engineering, National Taiwan Normal University | Sep 2022 - Jul 2024
- B.S. in Computer and Communication, National Pingtung University | Sep 2018 - Jun 2022
- Honors: RHCSA; FIRA Roboworld Cup First Place; National Technical Creativity Competition Third Place